

```
main | sudo tee /etc/apt/sources.list.d/ceph.list
```

Update your repository and install *ceph-deploy*, as shown below:

```
sudo apt-get update && sudo apt-get install ceph-deploy
```

A storage cluster

Creating a Ceph cluster: If at any point there is some problem, you can start over by executing the following commands:

```
ceph-deploy purgedata {ceph-node} [{ceph-node}]
ceph-deploy forgetkeys
```

Now on the admin node, you can use *Ceph-deploy* to perform the following.

Create a cluster:

```
ceph-deploy new {ceph-node}
ceph-deploy new ceph-node1
```

Install Ceph:

```
ceph-deploy install {ceph-node} [{ceph-node} ...]
ceph-deploy install ceph-node1 ceph-node2 ceph-node3
```

Add a Ceph mon:

```
ceph-deploy mon create {ceph-node}
ceph-deploy mon create ceph-node1
```

Gather keys:

```
ceph-deploy gatherkeys {ceph-node}
ceph-deploy gatherkeys ceph-node1
```

Add an OSD:

```
ssh ceph-node2
sudo mkdir /tmp/osd0
exit
```

Use *Ceph-deploy* on *admin* to prepare OSD:

```
ceph-deploy osd prepare {ceph-node}:/path/to/directory
ceph-deploy osd prepare ceph-node2:/tmp/osd0
```

Activate OSD:

```
ceph-deploy osd activate {ceph-node}:/path/to/directory
ceph-deploy osd activate ceph-node2:/tmp/osd0
```

Use *Ceph-deploy* to copy the configuration file and admin key to your admin node and your Ceph node, so that you can

use the Ceph CLI without having to specify the monitor address and *ceph.client.admin.keyring* each time you execute a command.

```
ceph-deploy admin {ceph-node}
ceph-deploy admin admin-node ceph-node1 ceph-node2 ceph-node3
```

Features of Ceph

▪ Fault tolerance

According to Wikipedia, “Fault-tolerance is the property that enables a system to continue operating properly in the event of the failure of (or one or more faults within) some of its components.” RADOS distributes objects across the storage cluster and replicates objects over the cluster, so in case of object loss over one of the OSDs, the rest of the OSDs, which are part of the cluster, will still have the copy of the object. Hence, there is no single point of failure.

▪ Scalability

As defined by Wikipedia, scalability is the ability of a system, network or process to handle a growing amount of work in a capable manner, or its ability to be enlarged to accommodate that growth. Ceph does not follow a centralised gateway method of monitoring data storing and sharing. The client directly interacts with the OSD, which in turn creates replicas of objects on each node, which ensures high availability of data.

Why Ceph?

There are many reasons for choosing Ceph over other distributed file systems like Hadoop.

- Ceph is an open source project written in C++ and C.
- Ceph provides a reliable scalable metadata server, which uses dynamic sub-tree partitioning with non-linear scaling.
- Since Ceph is designed to serve as a general-purpose file system, if it supported Hadoop workloads well, it could also be a general solution to other storage needs like providing big data solutions.
- Each file can specify its own striping strategy and object size. Flexible striping strategies and object sizes are important tuning parameters for Hadoop workloads. Ceph uniquely delivers object, block and file storage in one unified system. It is highly reliable, easy to manage and free. A Ceph node leverages commodity hardware and intelligent daemons, and a Ceph storage cluster accommodates large numbers of nodes, which communicate with each other to replicate and re-distribute data dynamically. A Ceph monitor can also be placed into a cluster of Ceph monitors to oversee the Ceph nodes in the Ceph storage cluster. 

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